



SENTIENT TOKEN AND ECONOMIC OPTIMIZATION FOR AGENTS

# Make agent optimization falsifiable.

An open operation-control and evidence layer that binds what an agent was asked to do, what actually ran, what it cost, and whether an external verifier accepted the result.

PUBLIC-GOODS REQUEST

**\$150,000**

Nine months. Three go/no-go tranches. No equity. Open methods, code, evidence, and negative results.

**809**

public GitHub stars

**24**

outside-authored repositories

**32 / 32**

official holdout verdicts

**OPEN**

MIT code and public proof

## PROBLEM

# The model call is not the economic unit.

Real cost and failure accumulate across the entire operation.

- Decomposition, parallelism, retries, tools, context, recovery, and verification can overwhelm a cheap model choice.
- A requested model can differ from the provider path that actually executes.
- Unknown local, subscription, setup, or human cost is often collapsed into zero.
- Model prose or stack status can be promoted to completion without an external artifact check.

**01 Declared plan**

Models, tools, topology, and limits

**02 Observed execution**

Calls, attempts, artifacts, and cost lenses

**03 External verdict**

Deterministic verifier outside the route

**04 Signed evidence**

Offline-reconstructable receipt chain

UNKNOWN STAYS UNKNOWN



WORKING SUBSTRATE

# Built before the grant.

Citadel starts with one command, then exposes stronger control only when the operation needs it.

- /do

Beginner path

Natural-language routing, verification, and a durable next action.
- PLAN

Portable contract

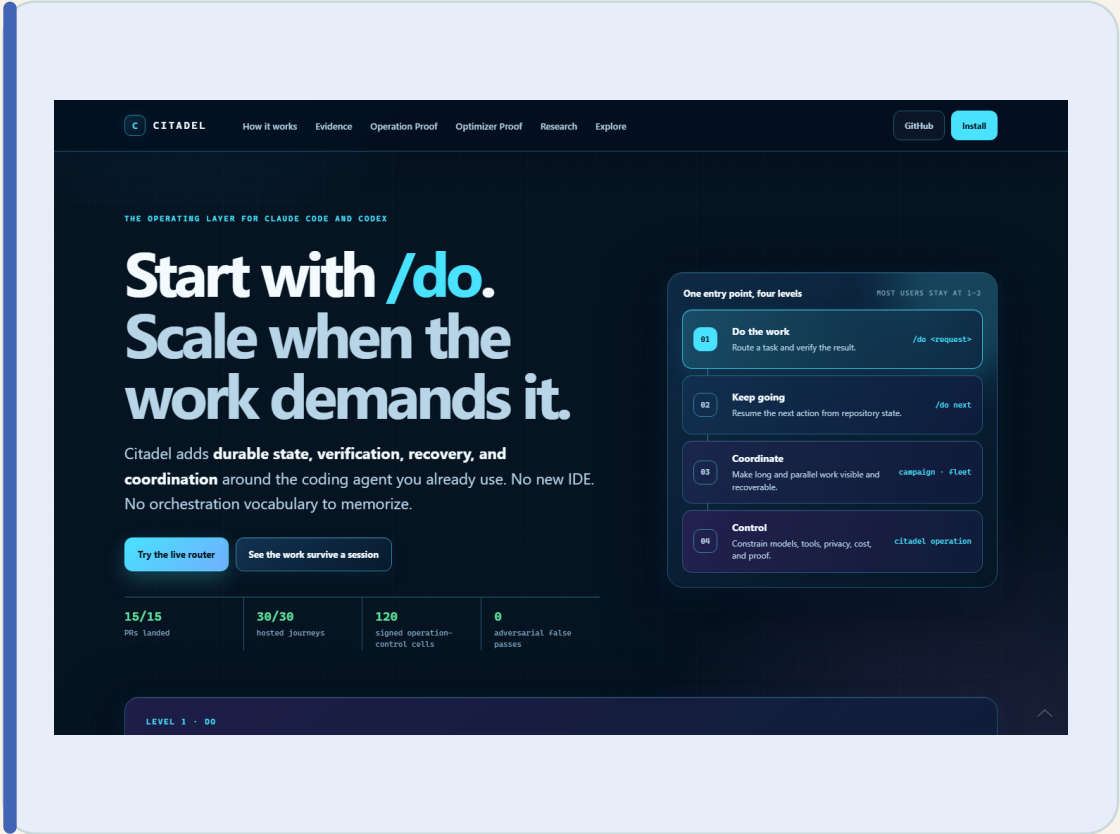
Models, tools, topology, retries, privacy, time, cost, and stop rules.
- RUN

Runtime coverage

Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- PROOF

Evidence plane

External artifact verdicts, failure preservation, signed receipts, and offline replay.



## OUTSIDE-AUTHORED REPOSITORY DIAGNOSTIC

# The evidence worked. The baseline did not.

Twenty-four distinct repositories. Routes sealed before evaluation. Every official verdict published, including the result that weakens the proposal.

**3 / 16**

sealed controller verified

**2 / 16**

direct Claude verified

**1 / 16**

local Qwen verified

**-1.26%**

comparison cost; not cash

**WHAT COMPLETED**

- Eight calibration and sixteen untouched evaluation tasks across four strata.
- Sixteen signed routes merged before model calls; 32/32 official verdicts.
- No task replacement, optional stopping, or positive-only publication.

**WHY THIS IS NOT A WIN**

Direct Claude passed only 12.5% overall and 0% in three strata. Matching or slightly exceeding that baseline does not establish useful quality preservation or general savings.

**BASELINE VALIDITY FAILED**



BUILT BEFORE FUNDING

# One maintainer already shipped the substrate.

Citadel is a public system with continuing repository pull, not a grant-funded mockup.

136

days public

518 / 542

main commits attributed to Seth

809 / 80

stars / forks

524

unique cloners in 14 days

WORKING SUBSTRATE

- Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- 49 workflows and 35 hooks across 29 lifecycle events.
- Windows, Linux, and macOS verification on every release path.

CLAIM DISCIPLINE

A synthetic support envelope passed 12/12 at 38.7% lower comparison cost. The outside-authored follow-up then invalidated its strong-baseline claim. Both remain public.

FUNDING SCALES A WORKING BASE

## NINE-MONTH RESEARCH AND ENGINEERING PROGRAM

# Funding buys a valid operation baseline.

Repair the retrieval and edit substrate exposed by outside-authored work, then rerun the economic question.

|    |                                    |                                                                        |              |
|----|------------------------------------|------------------------------------------------------------------------|--------------|
| 01 | <b>Valid workload and baseline</b> | Retrieval/edit contract, 60+ operations, complete cost boundary        | <b>\$30k</b> |
| 02 | <b>Adapter SDK</b>                 | Sentient ROMA plus two additional open stacks                          | <b>\$28k</b> |
| 03 | <b>Expected-value controller</b>   | Price verification, escalation, recovery, and failure                  | <b>\$35k</b> |
| 04 | <b>Prospective evaluation</b>      | Frozen multi-stack policy and statistical contract                     | <b>\$32k</b> |
| 05 | <b>Public proof and release</b>    | Hosted reconstruction, /do onboarding, support window, and maintenance | <b>\$25k</b> |



NO-GO RULES ARE PART OF THE PRODUCT

# Capped spend. Hard outcome gates.

COST BASIS

|                                       |          |
|---------------------------------------|----------|
| Maintainer labor                      | \$99,000 |
| Evaluation compute and tools          | \$27,000 |
| Hosted verification and retention     | \$7,000  |
| Reference hardware and measurement    | \$7,000  |
| Compatibility and accessibility costs | \$2,000  |
| Bounded contingency                   | \$8,000  |

**Total** **\$150,000**

PRIMARY PROSPECTIVE GATE

**> = 80%**

absolute verified completion

**> = 95%**

of valid frontier rate

**> = 30%**

lower measured end-to-end cost

**ZERO**

false passes and integrity failures

UNKNOWN COST CANNOT COUNT AS SAVINGS

## DIRECT FIT WITH A NAMED REQUEST FOR PRODUCTS

# An open optimizer that can prove when it is wrong.

## WHY FUND CITADEL

- Directly targets Sentient's Token and Economic Optimization request.
- Already binds a pinned Sentient ROMA stack without replacing its planner.
- A bounded synthetic pass plus a failed outside baseline identify exactly what funding must repair.

## WHAT STAYS PUBLIC

- MIT code, adapters, contracts, methods, verifiers, cells, and reports.
- No mandatory Citadel account, token, hosted evidence endpoint, or telemetry resale.
- Users keep project state, operations, receipts, and the right to run locally.

**Seth Gammon - solo builder and maintainer**

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Evidence: [github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR\\_START\\_HERE.md](https://github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR_START_HERE.md)

**OPEN. VERIFIABLE. FALSIFIABLE.**